An explicit constant for [OpenAI, *Uniform exclusion of Landau–Siegel zeros*] with H free: $c \geq 4.467 \times 10^{-3}$ (with Lu–Zaman–Zhao)

Joseph M. Reilly

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Version 1 (`note.md`, `note.tex`, `note.pdf`) is kept unchanged. This PDF is built from `docs/effectivity/note-v2.tex`, which follows `note-v2.md`. Computation: `src/effectivity.py`; output: `results/2026-10-20-effectivity/effectivity-output-v2.txt`. Comparison with Banks: `src/banks_compare.py`; output: `results/2026-10-20-paper-prep-2/banks-compare-output.txt`; table: `docs/effectivity/banks-comparison-v2.md`. Derivation: `docs/effectivity/derivation.md`. Sources: [S1]–[S10], listed at the end; [S1]–[S8] are as in `docs/effectivity/sources.md`, where the quotations are; [S10] is described in `refs/banks-2610.15483.md`.

Author's note

I am not a number theorist. This note was produced with AI assistance. Claude (Anthropic), working through Claude Code, wrote the derivation (`docs/effectivity/derivation.md`), the interval-arithmetic code in `src/`, and the comparison with Banks (`docs/effectivity/banks-comparison-v2.md`). A separate Claude session reviewed each stage and recomputed Banks's constants and the certified values by hand from the master inequality; that check is not independent of the model that produced the work. My role was to direct the project, decide what to check and what to claim, and review each stage before approving the next. Every inequality claimed here is certified in Arb ball arithmetic from exact rational parameters, and the scripts, tests and outputs that produce the numbers are in the repository. No specialist has reviewed the accounting. Until one does, the constants should be treated as unreviewed.

1 Background: Banks's explicit constant

[S1] proves that some absolute $c > 0$ exists. Its proof argues by contradiction along a sequence with $q \to \infty$ and $\delta := (1-\beta)\log q \to 0$, and leaves every analytic constant implicit.

W. D. Banks [S10] (arXiv:2610.15483, dated October 11, 2026) rearranges the argument as a direct proof with explicit constants:

- $\delta_0 = 81/(5888\, e^{3/2}) = 3.0696 \cdot 10^{-3}$ for $q \geq 10^{10}$ (his Theorem 1.1 and Remark 7.1);
- with Lu–Zaman–Zhao's computation, $(1-\beta)\log q \geq 3.0696 \cdot 10^{-3}$ for every $q \geq 3$. He states this rounded, as $3.0 \cdot 10^{-3}$.

Kerr (arXiv:2610.21099) and Green (hexagon:2610.00311v2, 2026-10-17) give no explicit constant.

This note uses the same master inequality as Banks's §7, with H left free and a few other changes (§4).

2 Statement

Assuming the proof in [S1] is correct (see §8), every real zero $\beta \in (0,1)$ of every primitive nonprincipal real Dirichlet L-function of conductor $q \geq 3$ satisfies

$$(1-\beta)\log q > 4.467 \times 10^{-3} \quad \text{using Lu–Zaman–Zhao's computation for } q \leq 10^{10} \text{ (the LZZ range),}$$

$$(1-\beta)\log q > 3.399 \times 10^{-3} \quad \text{using only Platt's computation for } q \leq 4 \cdot 10^5 \text{ (the Platt range).}$$

The argument of [S1] alone, without any computation for small q, gives $(1-\beta)\log q > 5.04 \times 10^{-4}$ for every $q \geq 3$ except $q = 8$. These bounds are certified in Arb ball arithmetic from exact rational parameters.

3 Explicit inputs

Definitions. Here $M = N^4$ and $U = N^{4/3}$, with $L = \log U$. Banks writes N^4 for our M and uses M for our $U = N^{4/3}$. The real variable of the L-function is σ throughout. derivation.md writes s for it; s is kept for the shift in `docs/effectivity/derivation-banks-device.md`.

Input	Statement used	Source and range
Hadamard identity (2.3)	$\mathrm{Re}(-L'/L)(\sigma,\chi) = \frac{1}{2}\log(q/\pi) + \frac{1}{2}\psi((\sigma+\varepsilon)/2) - \sum_\rho \mathrm{Re}\,\frac{1}{\sigma-\rho}$, terms ≥ 0, real $\sigma > 1$	as stated in [S1] §2; checked against Davenport, 2nd ed., Ch. 12 (17)–(18), p. 83
ζ log-derivative	$-\zeta'/\zeta(\sigma) < 1/(\sigma-1)$, $\sigma > 1$	[S6] Lemma 12; final elementary inequality proved here for all $\sigma > 1$ (§4)
Digamma	$\psi((\sigma+\varepsilon)/2) \leq \psi(3/2) = 2 - \gamma - 2\log 2$ for $1 < \sigma \leq 2$	ψ increasing
Mertens	$\log x + E - \frac{1}{2\log x} < \sum_{p \leq x} \frac{\log p}{p}$ $(x > 1)$; $\sum_{p \leq x} \frac{\log p}{p} < \log x + E + \frac{1}{2\log x}$ $(x \geq 319)$	Rosser–Schoenfeld Thm 6, (3.21)–(3.22), p. 70, read in the original
Chebyshev	$\theta(x) < (1 + 7.5 \cdot 10^{-7})x$, $x > 0$	[S5] Platt–Trudgian Cor. 2 (arXiv numbering)
Primes dividing q	$\sum_{p \mid q} \frac{\log p}{p} \leq \min\left(\frac{1}{2}, \frac{\log 2}{2\log q} + \frac{1}{3}\right)\log q$	elementary
Lemma 6	$S_2 \leq 192\, MH^{-1/3}U$, $S_1 \geq MH^{2/3}U/(4 \cdot 97^2)$, given $N \geq H$ and $H^{2/3}N^{4/3} \geq 2 \cdot 97^2$	[S1] lines 517–550 (= Banks Lemma 4.2)
Small q	no real zero with $\sigma > 0$ for $q \leq 4 \cdot 10^5$ ([S9]); $\delta > 1/5$ for $q \leq 10^{10}$	[S9] Thms 10.1–10.2, read in the original; [S2] Theorem 1.1

4 The master inequality, the parameters, and the comparison with Banks

Combining the explicit Lemma 2, (5.1), (5.6) with the θ bound, and Lemma 6 (derivation.md §§1–5): if a real zero with $\delta = (1-\beta)\log q$ exists and $N \geq H$, then

$$1 \leq \tfrac{3}{4}(1+\rho_H) + \frac{C_\ell + \frac{1}{2}(1+\rho_H)}{\lambda} + \frac{e}{2}\,\delta\lambda + \frac{C_H + (1+\rho_H)\log 8}{L} + \frac{(1+7.5\cdot 10^{-7})\cdot 4\cdot 97^2}{H^{2/3}L} + \frac{6\cdot 97^2}{H^{2/3}U},$$

where $\rho_H = 7{,}226{,}112/H$, $C_\ell = e/4 + c_q$, $C_H = \log H + \frac{1}{2\log H} + \frac{1}{2L} + \frac{e}{2}\kappa$, $\kappa = \frac{1}{2}\psi(3/2) - \frac{1}{2}\log\pi = -0.55412\ldots < 0$, $L = \log U = \frac{4}{3}\log N$ and $\lambda = L/\log q$. This is Banks's Step 3 inequality (§7), term for term, in a different accounting (`banks-comparison-v2.md`, rows 1–21).

Take $N = \lceil q^\gamma \rceil$. For all $q \geq q_0$:

- $\lambda \in [4\gamma/3,\ 4\gamma/3 + \frac{4}{3}\log(1+q_0^{-\gamma})/\log q_0]$;
- every term except the δ term is largest at $q = q_0$;
- so every real zero with $q \geq q_0$ has $\delta > \delta_0(q_0, \gamma, H)$, with δ_0 given in closed form (function `delta0` in `src/effectivity.py`).

For fixed (γ, H), δ_0 is determined exactly; no fixed split of the margin is used. The parameters were found by a floating-point search, then certified in Arb at the exact values below (every printed bound is the Arb lower endpoint rounded down).

Range	q_0	γ	H	c_q	certified $\delta_0 >$
Platt	400,001	201/10	1,995,262,314	$\frac{\log 2}{2\ell} + \frac{1}{3}$	$\mathbf{3.3992904026 \times 10^{-3}}$
Lu–Zaman–Zhao	$10^{10}+1$	153/10	2,818,382,931	$\frac{\log 2}{2\ell} + \frac{1}{3}$	$\mathbf{4.4673780338 \times 10^{-3}}$
argument alone	3	671/5	1,122,018,454	$\frac{1}{2}$	$\mathbf{5.0425352961 \times 10^{-4}}$

Unlike Banks's §7, our rows do not need $H \leq q$: the Platt row has $H > q_0$. Banks uses $H \leq q$ only to get $N \geq H$, which γ secures here (side condition $N \geq H$ below).

From Banks's $3.0696\cdot 10^{-3}$ **to** $4.467\cdot 10^{-3}$ (COMPUTED; `banks-comparison-v2.md` (c)). The waterfall below starts from Banks's published value at $q_0 = 10^{10}$ and switches his choices to ours one at a time. γ is re-optimized at each step; H is the paper's until the last step. The factors depend on the order.

Change from Banks's choice to ours	Factor
exact solve instead of the 3/32 split and the $\log q > 23$ absorption	$\times 1.024$
tight upper end of λ ($N \leq q^\gamma + 1$, not $q^{\omega+1}$)	$\times 1.046$
our S_1, S_2 bounds ($S_1 \geq N^4 H^{2/3} U/(4\cdot 97^2)$, not $N^4 U$)	$\times 1.044$
Platt–Trudgian θ bound instead of $\theta(M) \leq 2M$	$\times 1.003$
$\kappa < 0$ kept (with the Mertens terms)	$\times 1.02$
shift 1 in $\sigma = 1 + s/\log X$, instead of Banks's 3/2 (a cost)	$\times 0.88$
our primes-dividing-q bound, instead of Banks's $\frac{1}{2}S_0$ device (a cost)	$\times 0.86$
H free ($\gamma = 153/10$, $H = 2{,}818{,}382{,}931$), instead of $H = 86{,}713{,}344$	$\times 1.68$

Check (COMPUTED, `results/2026-10-20-paper-prep-3/note-v2-factor-check.txt`): the exact factors multiply to the ratio $4.4673780338/3.0695555319 = 1.45538$. The seven rounded factors other than θ multiply to 1.45018 (-0.36%), and with θ to 1.45453 (-0.06%); both lie within

the rounding of the stated factors. At the paper's own $H = 86{,}713{,}344$, the same three ranges give 2.064, 2.652 and 0.3245 ($\times 10^{-3}$). A further refinement of the accounting is under review and is not used here.

Side conditions (derivation.md §7), certified at q_0 and implied for all $q \geq q_0$: $H \geq 319$; $N \geq H$ ($\gamma \log q_0 > \log H$); $H^{2/3}N^{4/3} \geq 2 \cdot 97^2$ (from $N \geq H \geq 319$); $U \geq 3$, and $\sigma = 1 + 1/L \in (1, 2]$; $\kappa < 0$.

Lemma 12's last step. $\varphi(\sigma) = \frac{1}{2\sigma-1} - \left[\frac{1}{\sigma-1} - \frac{1-2^{-\sigma}}{\sigma} - \frac{3^{1-\sigma}}{\sigma-1}\right] > 0$ for all $\sigma > 1$: on $(1, 1.01]$ by the exact chain $(1 - 3^{-t})/t \leq \log 3$, $(1 - 2^{-\sigma})/\sigma \geq 1/(2\sigma)$, reducing it to $\log 3 - 1/2.02 < 1/1.02$; on $[1.01, 5]$ by Arb on 4,000 intervals (minimum certified lower bound of φ: 0.03878); for $\sigma \geq 5$ by $\varphi > 0 \Leftarrow \sigma^2 - 5\sigma + 2 > 0$.

5 Remarks

- **Banks's numbers recomputed.** Each of the 12 numeric claims in Banks's Lemma 2.1 and §7 was recomputed independently in Arb, and all 12 hold. His $\delta_0 = 81/(5888e^{3/2})$ recomputes as $[0.00306955553193322 \pm 2.9 \cdot 10^{-18}]$ (`src/banks_compare.py`, Part A).
- **Lemma 3.** Our Lean proof of [S1]'s Lemma 3 (`lean-checks/Lemma3.lean`, theorem `Lemma3.interpolation`) has hypotheses identical to Banks's condition (3.1), the hypothesis of his Proposition 3.4 (clause-by-clause comparison: `docs/packet-related-work-draft-v2.md`, which points to the table in `docs/packet-related-work-draft.md`).

6 Small conductors

- **Platt range:** for $q \leq 4 \cdot 10^5$ there is no real zero with $\sigma > 0$ at all, so the bound holds vacuously. This range includes $q = 8$, the conductor the paper's field construction excludes ($d = 2$). $q > 4 \cdot 10^5$ is covered by the first row.
- **Lu–Zaman–Zhao range:** for $q \leq 10^{10}$, $\delta > 1/5$, which also covers $q = 8$; $q > 10^{10}$ is covered by the second row; so $c = \min(4.467 \cdot 10^{-3}, 1/5) = 4.467 \cdot 10^{-3}$.
- **Argument alone:** covers $q \geq 3$ with $q \neq 8$ by taking γ large ($\lambda \approx 179$), i.e. $N \approx q^{134}$. $q = 8$ needs a separate input.

7 The final constant

$$c = 4.467 \times 10^{-3} \text{ (with Lu–Zaman–Zhao)}, \qquad c = 3.399 \times 10^{-3} \text{ (with Platt only)},$$

both rounded down from the certified values.

Context.

- The companion paper on the zero-free half-plane $\operatorname{Re} s > 7/8$ would, if correct, give $1 - \beta \geq 1/8$ for every such β, a far stronger statement.
- A real zero with $\delta \approx 0.003$ is far outside anything ever observed. These numbers are for checking the argument, not a claim about zeros.

Comparison with an earlier version (docs/explicit-constant.md, written 2026-10-08, public in reviewer packet v1 on 2026-10-09: tag reviewer-packet-v1, commit 3a36016, Zenodo 10.5281/zenodo.23269528). Its figures were $3.0379 \cdot 10^{-4}$ (row "all $q \geq 3$"), $1.99806 \cdot 10^{-3}$ ($q > 4 \cdot 10^5$) and $2.58930 \cdot 10^{-3}$ ($q > 10^{10}$):

- **Same inputs reproduce it.** With its inputs switched back on, this code gives slightly larger maxima ($3.040 \cdot 10^{-4}$, $2.0000 \cdot 10^{-3}$, $2.5918 \cdot 10^{-3}$, rounded down). The earlier note reported a δ with a positive safety margin ($F \approx 9.4 \cdot 10^{-5}$) rather than the maximum.
- **"All $q \geq 3$" is used there twice:** its table row "all $q \geq 3$" ($3.0379 \cdot 10^{-4}$) covers small q the same way as row 3 here: by taking λ large, not by any computation ($q = 8$ again excepted). Its bold conditional statement "$(1-\beta)\log q \geq 0.00258\ldots$ with $q \geq 3$" combines the $q > 10^{10}$ row with the small-q computations of Lu–Zaman–Zhao (and Platt). Source: `results/2026-10-20-paper-prep-4/task2-earlier-constant.txt`.
- **It used three inputs replaced here:** an unsourced "$\sum_{p \leq x} \log p/p < \log x$" for the primes up to H, which costs $|E| = 1.33$ in C_H against the sourced Rosser–Schoenfeld upper bound; Rosser–Schoenfeld Thm 9 for θ; a fixed H.
- **The gain comes mainly from freeing** H ($\times 1.6$). Of the other changes, $c_q = \frac{1}{2}$ is worse than its c_q at large q but better at $q = 3$, and the sourced Mertens bound and the θ bound are improvements.

8 Assumptions (everything the constant depends on)

1. **The paper's proof is correct:** in particular Lemma 3, Corollary 4, Lemma 6, Lemma 7 and the bounds (5.1) and (5.6).
   - This note does not re-verify them mathematically.
   - Independent evidence: the Lean formalization of Theorem 1 is accepted by the official comparator with two kernels (Lean and nanoda), including a version rerouted through the paper's own Lemma 3. See the replay packet (DOI 10.5281/zenodo.23269527; `grwtsk/openai-math` #40) and the statement audit on `grwtsk/openai-math` #29 (`https://github.com/grwtsk/openai-math/issues/29#issuecomment-6296359378`).
   - The Lean proof has existential constants only and does not contain these numbers.
2. **The Hadamard identity (2.3).** Checked against Davenport, *Multiplicative Number Theory*, 2nd ed., Ch. 12, (17) and (18) on p. 83, with $a = \varepsilon$ (p. 70). Taking real parts of (17) and using (18), $\operatorname{Re} B(\chi) = -\sum \operatorname{Re} 1/\rho$, gives (2.3).
3. **Rosser–Schoenfeld Theorem 6,** read in the original: (3.21) for $1 < x$ and (3.22) for $319 \leq x$, p. 70. $E \approx -1.33258\,22757\,33221$, (2.11). This agrees with the quotation in arXiv:1703.08032 and with the enclosure in `src/effectivity.py`.
4. **Platt (2016) [S9],** read in the original. Theorem 10.1 states GRH for primitive characters of modulus $q \leq 400\,000$, up to a positive height. Theorem 10.2 states $L(1/2, \chi) \neq 0$ for $q \leq 2\,000\,000$. Together (our inference) they rule out a real zero in $(0, 1)$ for $q \leq 4 \cdot 10^5$, matching the Lu–Zaman–Zhao summary. Needed only for the $3.399 \cdot 10^{-3}$ statement.
5. **Lu–Zaman–Zhao Theorem 1.1** (arXiv:2602.03626v1, a preprint), read in the original. Needed only for the $4.467 \cdot 10^{-3}$ statement.
6. **Platt–Trudgian Corollary 2,** read in the arXiv version (arXiv:1407.1914, v1, the only version; sha256 in `docs/effectivity/sources.md`). The published version [S5] was not available to check whether the corollary has the same number there.
7. **Lemma 12 of arXiv:2511.13898** (a preprint), read in the original. Its reduction steps are taken from that proof; its final inequality is proved here.

8. $\chi(p) = (d/p)$ **for odd** $p \nmid q$**, and** $|d| \le q$**,** as stated in the paper.
Not used: any unproved hypothesis (GRH or similar), or any floating-point value in a claimed inequality.

References

[S1] OpenAI, *Uniform exclusion of Landau–Siegel zeros*, October 1, 2026, `github.com/openai/math`, commit adc7f12, family 003.

[S2] R. F. Lu, A. Zaman, H. Zhao, *Numerical computations concerning Landau–Siegel zeros*, arXiv:2602.03626v1 (2026).

[S3] J. B. Rosser, L. Schoenfeld, *Approximate formulas for some functions of prime numbers*, Illinois J. Math. 6 (1962), 64–94, `doi:10.1215/ijm/1255631807`.

[S4] C. Axler, *New estimates for some functions defined over primes*, arXiv:1703.08032.

[S5] D. J. Platt, T. S. Trudgian, *On the first sign change of* $\theta(x) - x$, Math. Comp. 85 (2016), 1539–1547, `doi:10.1090/mcom/3021`; arXiv:1407.1914. Corollary 2 is cited from the arXiv version (v1), the version read. Its numbering in the published version was not confirmed.

[S6] T. Zhao, *Upper bounds on gaps between zeros of L-functions*, arXiv:2511.13898, Appendix, Lemma 12.

[S7] H. Davenport, *Multiplicative Number Theory*, 2nd ed., rev. H. L. Montgomery, Springer GTM 74.

[S8] Elementary facts (no citation): ψ increasing on $(0, \infty)$, $\psi(3/2) = 2 - \gamma - 2\log 2$, and $\log p/p \le (\log p)/2$ for $p \ge 2$ (`docs/effectivity/sources.md`, S8).

[S9] D. J. Platt, *Numerical computations concerning the GRH*, Math. Comp. 85 (2016), 3009–3027, `doi:10.1090/mcom/3077`.

[S10] W. D. Banks, *On the OpenAI proof that Landau–Siegel zeros do not exist*, arXiv:2610.15483v1 (2026).